

Project Initialization and Planning Phase

Date	15 March 2024
Team ID	XXXX
Project Title	AI Learning Support Assistant
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposal report aims to transform loan approval using machine learning, boosting efficiency and accuracy. It tackles system inefficiencies, promising better operations, reduced risks, and happier customers. Key features include a machine learning-based credit model and real-time decision-making.

Project Overview	
Objective	The primary objective of the AI Learning Support Assistant is to help students understand difficult academic concepts by combining Natural Language Processing (NLP) , Deep Learning , and Generative AI . The system detects the user's emotional state from their learning query and provides personalized learning guidance along with analytics and interaction history
Scope	The project focuses on developing a web-based learning assistant that: • Accepts learning problems from students. • Predicts emotions using BiLSTM and BERT models. • Detects primary and secondary emotions. • Generates personalized learning guidance using the Google Gemini API. • Stores previous learning interactions in a CSV file. • Displays analytics dashboards using Plotly. • Allows users to download their interaction history. The application is intended for students and self-learners who need intelligent academic assistance during their learning process.
Problem Statement	
Description	Students often struggle to understand difficult academic concepts and may experience confusion, frustration, or anxiety while studying. Traditional learning resources provide the same explanation to every learner and do not consider the learner's emotional state. Additionally, students often lack a simple way to review previous learning

	interactions or monitor their learning progress.
Impact	The proposed solution helps students by: • Providing personalized academic guidance. • Identifying emotions associated with learning difficulties. • Improving understanding of challenging topics. • Maintaining previous learning interactions for future reference. • Visualizing learning trends through analytics dashboards.
Proposed Solution	
Approach	The application follows these steps: • The student selects a subject and enters a learning problem. • The input text is processed and analyzed. • The BiLSTM model predicts emotion probabilities. • The BERT model performs emotion classification. • The Mixed Emotion Detector identifies primary and secondary emotions. • The detected emotion and learning problem are sent to the Google Gemini API to generate personalized learning guidance. • The interaction is stored in a CSV file. The application displays emotion analysis, AI guidance, previous interaction history, and analytics dashboards
Key Features	• Emotion prediction using BiLSTM • Emotion prediction using BERT • Mixed emotion detection • Personalized AI learning guidance using Google Gemini API • Emotion confidence analysis • Interactive BiLSTM probability chart • Previous interaction history • Download interaction history (CSV) • Analytics dashboard with Plotly • Emotion distribution visualization • Confidence trend visualization • Subject-wise analytics • Responsive Streamlit interface

Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	CPU for model inference	Intel Core i5 / Ryzen 5 or equivalent
Memory	RAM	Minimum 8 GB
Storage	Project files, trained models, logs	Minimum 5 GB available storage
Software		

Software

Resource Type	Description	Specification / Allocation
Framework	Web Application	Streamlit
Programming Language	Backend Development	Python 3.11
Deep Learning Framework	Model Development	TensorFlow
NLP Framework	Transformer Model	Hugging Face Transformers
Libraries	Data Processing & Analytics	Pandas, NumPy, Scikit-learn, Plotly
Development Environment	IDE	Visual Studio Code

Data		
Data		
Resource Type	Description	Specification / Allocation
Emotion Dataset	Dataset used for model training	Multi-class emotion dataset (Joy, Love, Anger, Fear, Sadness, Surprise)
User Input	Learning problems entered by students	Text input
Interaction History	Stored prediction records	CSV format (interaction_history.csv)